

Yang Zhenzhao

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Education

National University of Singapore

Aug 2024 – Present

Bachelor of Engineering in Computer Engineering

Hwa Chong Institution (High School, A'Level)

Jan 2016 – Dec 2021

Experiences

PSA International

May 2026 – Present

Data Engineer Intern

- Designing and implementing **data pipelines**, **ETL processes**, and data warehousing solutions on **Databricks** to support PSA's next-generation data analytics platform across port and business operations.
- Developing efficient and scalable data pipelines, ensuring **data quality and integrity** to meet business objectives, while collaborating with cross-functional teams to translate data requirements into pipeline design.
- Building interactive dashboards with **Power BI** to communicate data analysis effectively to various stakeholders, and supporting optimisation of existing data platform infrastructure.

Massivue

May 2025 – Aug 2025

AI Engineer Intern

- Built an AI-powered process workflow analysis platform integrating **process mining**, real-time analytics, and **RAG-enhanced insights** for operational optimisation.
- Implemented a dual AI architecture with seamless switching between cloud-based **GPT-4** and a locally compressed **Gemma-3B** model optimised for on-premise deployment.
- Developed interactive dashboards (**React/TypeScript, Node.js**) to surface bottlenecks and drive measurable workflow improvements.

Amcor Flexibles

Feb 2024 – May 2024

Data Automation Intern

- Developed **Excel-based automation tools** to streamline HR workflows, including certification expiry tracking and automated alerts.
- Automated payroll-related processes using **VBA**, improving data accuracy and operational efficiency.
- Collaborated with HR and IT teams to reduce manual workload and improve compliance monitoring.

Singapore Police Force

Mar 2022 – Feb 2024

Full-time National Serviceman (Mandatory national service for all male Singaporean citizens)

- Special Operations Command, K9 Unit.

Notable Projects (please visit my Github to view all projects/hackathons)

PSA Code Sprint (3rd runner up – Top 4 out of 400+ teams, best team for

Oct 2025

Problem Statement 3: Redefining Level 2 Product Ops)

Machine Learning, Full-stack Web Development

- Part of a team of 4 that created an intelligent diagnostic platform for PSA Singapore's port operations, automating Level 2 support ticket analysis and resolution.

Real-time Fall Detection System on STM32

Aug 2025 – Sep 2025

Computer Organisation

- Built a real-time fall detection system on the STM32 B-L4S5I-IOT01A board, fusing accelerometer, gyroscope, and barometer data with a moving-average filter implemented in **ARM assembly**.
- Integrated on-board buzzer and LED alerting with a Wi-Fi to ntfy.sh to Python bridge to push notifications to Telegram, completing the embedded-to-cloud pipeline end-to-end.

NUS LifeHack (Finalist – Top 10 out of 60+ teams, most impressive usage of data award, best team for Theme 1: Health and Well-being)

July 2025

Full-stack Mobile Development

- Part of a team of 4 to build an app that integrates real-time Singapore health data, GPT-powered health guidance, and community reporting to build an AI-driven, multilingual public health ecosystem.

DSTA BrainHack Code Exp (Finalist – Top 20 out of 80+ teams)

June 2025

Full-stack Mobile Development

- Part of a team of 5 to develop an app that uses AI and community insights to detect online scams, analyse suspicious URLs, and educate users through interactive modules, analytics, and a scam discussion forum.

SMF & NUS Maritime Hackathon (Called for top 3 in University Category out of 93 teams)

Jan 2025

Machine Learning

- Part of a team of 4 to develop and train a machine learning model with over 90% accuracy to classify ship repair severity levels based on textual annotations, using a dataset of 10,000+ ship entries.

Clubs & Societies

NUS Computing Technopreneurship Society (StartIT)

Aug 2024 – May 2025

Project Manager

- Plan and execute **BitHacks**, the flagship startathon by StartIT.
- Plan and host fireside chats with tech startup co-founders.

NUS Social Impact Catalyst

Aug 2024 – May 2025

General Member

- Communicated with guest mentors and startup groups within the society to brainstorm ideas, craft pitches.

Relevant Skills/Tools

- **Programming Languages:** C, C++, Python, Java, JavaScript, TypeScript, SQL
- **Frameworks & Libraries:** React.js, React Native, Node.js, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas
- **Embedded & Digital Systems:** Arduino, Raspberry Pi, Esp32, MCXC444, FPGA (Vivado)
- **Data Analysis & Visualization:** Excel, VBA, Power BI, MySQL, exploratory data analysis
- **Tools & Platforms:** Git, Fusion 360, Cursor, Figma
- **Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Real-Time Operating System, Signals and Systems, Computer Organisation, Transistor-level Digital Circuits, Embedded Systems & Sensor Integration